

Resident Services & 311 Virtual Agent

FROM \$8 A CALL TO \$0.10

An always-on, multilingual agent that answers resident questions, files and tracks 311 service requests, and routes only the exceptions to staff — grounded in your knowledge base, with every answer cited and every action logged.

THE PROBLEM

- The same FAQs — trash days, permits, potholes, payments — consume scarce agent time.
- Live channels cost roughly \$8 a contact; self-service is about a dime.
- After-hours and language gaps push residents to call back, inflating volume.
- Staff copy-paste between systems instead of resolving the resident's actual issue.

\$8.01 per live-agent contact — ~80× the cost of self-service (Gartner, 2019)

98,028 / day contact volume at a single big-city 311 (NYC OTI, 2023)

THE GOVERNED SOLUTION

1 Understand the request

2 Ground the answer

3 Take the safe action

 **Human gate** (*human gate*)

5 Log & learn

Read and file-a-request are automated; commitments are not. The agent drafts; a person commits.

THE ROI

2,700 inquiries handled in month one, North Carolina DES assistant

~20% of all 311 interactions handled by Denver's 'Sunny' assistant

45% peak-season live-agent staffing reduction, Denver

Executive case: Every contact the agent resolves is roughly an \$8 live interaction turned into a dime — at 311 volumes that is a budget line, not a rounding error. Productivity research shows the biggest lift goes to your newest agents, shortening ramp time and turnover cost.

SECURITY & COMPLIANCE

Identity & least privilege — Cognito + custom:slg_role; deny-by-default MCP gateway; effective scope = agent grant $\cap$ user entitlement.

Data protection — PII masking before inference; KMS encryption; S3 Object Lock (WORM) audit; private VPC + Bedrock VPC endpoint.

AI guardrails & HITL — Bedrock Guardrails; grounded, cited answers; consequential action withheld in code until a human approves.

Compliance posture — Bedrock FedRAMP High + DoD IL-4/5 in GovCloud; CJIS / IRS-1075 alignment; CloudTrail; ADA WCAG 2.1 AA.

Deploy: Standalone (own VPC/edge/identity/data/WORM audit) or + Whole-of-Government layer. Full CloudFormation + Terraform parity, commercial & GovCloud (US).